

THE EQUITY EXPRESSION OF THE AI CAPEX THESIS

The Toll Road Problem

July was the worst month on record for semiconductor stocks — a 21% fall in the Philadelphia Semiconductor Index, its steepest since October 2008 — and almost none of it was caused by bad results. Companies reported records and were sold anyway. Tonight SpaceX, the year's largest new listing and now an AI-infrastructure business as much as a rocket company, reports audited earnings for the first time. The question the market is asking has quietly changed, and understanding that shift is the whole story.

Method note: external market data is drawn from public reporting and attributed with as-of dates; internal views come from the Heyokha Wiki's *AI Capex* theme page (last reviewed 4 August 2026) and are labelled as views, not verified forecasts.

A reframe worth holding onto

For two years the AI debate was about *supply* — enough chips, enough power, enough memory bandwidth. The internal Heyokha view is that this question has largely been answered in the affirmative, and that the market is now fighting the wrong battle. The unresolved question is not whether the infrastructure can run profitably. It is whether enough people want to pay for what it produces.

Imagine a city that borrows heavily to build a twelve-lane motorway across its centre. The engineering is superb: the surface is smooth, the gradients are gentle, the tolling gantries work flawlessly. Two years in, the city discovers something awkward. The road is magnificent, and traffic is real — but when surveyed, half the residents say they would give up using it for a month for the price of a sandwich. The motorway was never the risk. The toll revenue was.

THE STORY IN 60 SECONDS

–21%

The Philadelphia Semiconductor Index in July 2026 — its worst month since October 2008, and more than 28% below its 22 June record close. On nearly half of July's sessions the index moved at least 4%. *Fortune*, as of 31 July 2026.

–\$5.9bn

Alphabet's Q2 2026 free cash flow — negative for the first time since its 2004 listing — as quarterly capex roughly doubled to \$44.9bn. Full-year guidance was raised to \$195–205bn. *Company reporting*, Q2 2026.

~\$1.7tn

SpaceX's market value ahead of tonight's first-ever earnings report, with the shares at \$114.53 on 3 August — down roughly half from a 16 June high of \$225.64. *Investing.com / CNBC*, as of 3 August 2026.

WHAT ACTUALLY HAPPENED IN JULY

Records got sold

The mechanics of the July selloff are worth walking through slowly, because they are genuinely counterintuitive. This was not a quarter of disappointing numbers. Samsung guided to quarterly operating profit up roughly nineteen times year-on-year and the shares fell about 7%, erasing something like \$80bn of market value on a record result. TSMC reported roughly \$40bn of quarterly revenue at about 68% gross margin, guided higher, and raised capex — and fell. ASML, which the internal Heyokha research describes as "the least replaceable company on Earth," raised its 2026 forecast, reported strong orders and announced around 30% capacity expansion. It fell too.

The equipment makers took the worst of it: Applied Materials and Lam Research each lost roughly 30% over the month, KLA close to 40% at one point. These are, in the words of the internal note, companies "quite literally printing money" — but their order books depend on new fabrication plants being built, which is a bet on the *next* round of capex, not this one.

So what changed? The internal Heyokha view, drawn from White Box's 4 August issue, identifies five pressure points arriving more or less at once. A report that Meta was considering selling data-centre capacity to outside customers punctured the compute-scarcity narrative (Meta later denied it). Samsung's record-but-not-accelerating profits showed how far expectations had run. TSMC's and ASML's investment plans were suddenly read as future *supply* rather than evidence of demand. Alphabet's and Amazon's capex increases stopped being automatically welcome once free cash flow turned negative. And NVIDIA's willingness to backstop its own customers' financing made the demand picture look, in the note's phrase, "incestuous."

THE ESSENTIAL DISTINCTION

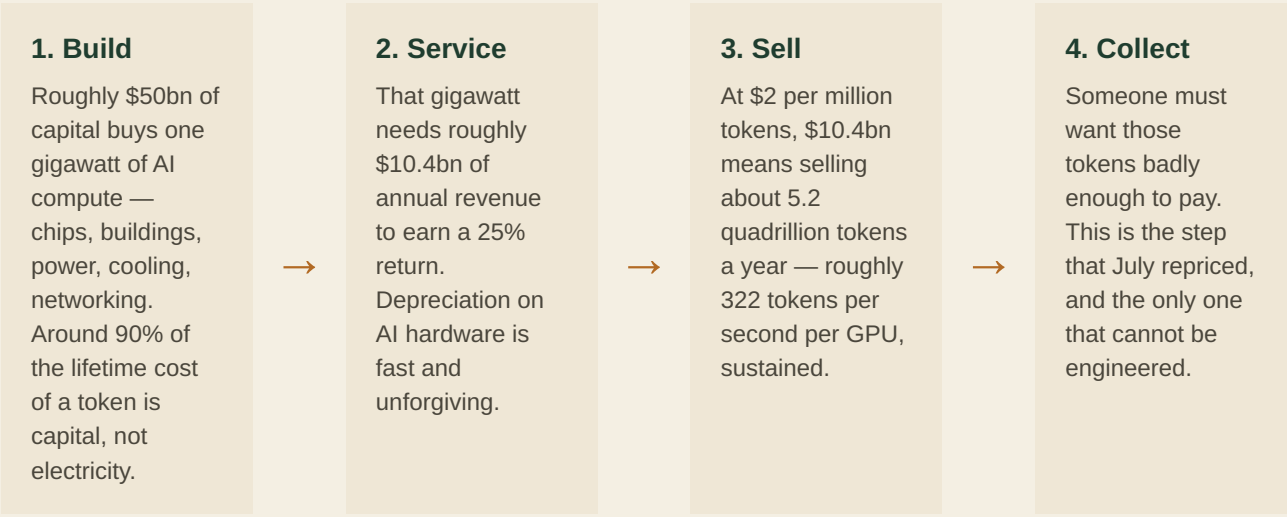
A rotation is not a crash. The S&P 500 sat roughly 1% from an all-time high while semiconductors fell 21%. Investors did not leave the market in July. They left one trade inside it.

Two mechanical amplifiers turned an ordinary correction into a historic one. Internal research documents a large AI-focused fund running roughly \$45bn at four-to-five times leverage with essentially no hedging; when prices fell, prime brokers margin-called and the fund lost 67% in a single month. Separately, two-times leveraged single-stock ETFs tied to Samsung and SK Hynix, launched in Korea at end-May just as the memory rally peaked, saw assets fall from around \$50bn to \$17bn by late July, with Citi estimating roughly \$39bn of retail losses. These figures are the provider's reporting and we have not independently verified them. The consolation, per the same note: "the casino has been partially cleared out," so semiconductor companies can go back to being judged on orders, utilisation and returns on capital.

THE MECHANISM

How a gigawatt has to pay for itself

To see why the demand question is now the binding one, it helps to follow a single unit of AI infrastructure from cheque to cash flow. The internal research puts the current build cost at roughly \$50bn per gigawatt of compute — and notes NVIDIA's own projection that this figure could double to \$100bn.



Gigawatt economics per Heyokha Wiki, *AI Capex* — internal view sourced to White Box (9 July 2026). Not independently verified.

Scale that arithmetic to the whole cycle and you get the number that matters. Assume \$2–3tn of incremental AI-related invested capital — which the internal research calls "almost definitely a lower-end estimate." A normal 10–15% after-tax return on that capital requires \$200–450bn of annual *operating profit*. At a 40% operating margin, that implies \$500bn to \$1.1tn of recurring annual revenue.

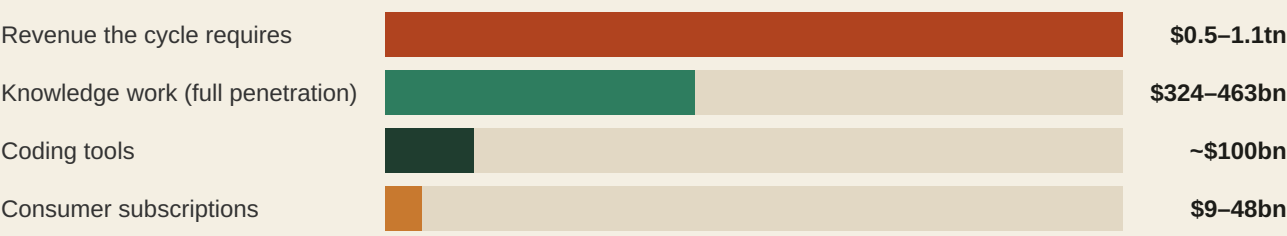


Return roadmap: Heyokha Wiki, *AI Capex*, internal view sourced to White Box (4 August 2026). Illustrative arithmetic, not a forecast. Hyperscaler operating margins are estimated at 10–20% today, which makes the 40% assumption generous.

THE DEMAND SIDE

How big is the market for intelligence?

Here the internal research does something unusual and useful: rather than asserting that demand is or is not there, it sizes each addressable segment and compares the total against the requirement above.



Segment sizing per Heyokha Wiki, *AI Capex* — internal view sourced to White Box (4 August 2026). Bars are scaled illustratively to the upper bound of the requirement. Not independently verified; not a forecast.

Coding is the strongest market and, on the internal read, "the vast majority of current revenue." Against roughly 36.5 million professional developers worldwide, Microsoft entered 2026 with 4.7 million paid GitHub Copilot subscribers, before Cursor, Claude Code or Codex. At \$50, \$100 or \$200 per developer per month, full penetration yields \$22bn, \$44bn or \$88bn a year. The internal conclusion is blunt: coding "is certainly not a one-trillion-dollar-a-year business right now."

Consumer subscriptions are smaller still. ChatGPT reportedly has more than 900 million weekly users but around 50 million paying subscribers — near 5.6% conversion, meaning roughly 94% choose the free product. Fifty million at \$20 a month is about \$12bn a year, already one of the largest consumer subscription businesses on Earth. Double the users and nearly double conversion to 10% and you reach roughly \$43bn. As the internal note puts it, that would be "an extraordinary business. But it would not be an infrastructure-sized business."

Then there is the study that gives the bear case its sharpest edge. Stanford's Digital Economy Lab asked Americans in April 2026 what they would need to be paid to give up generative-AI chatbots for a month. The mean was \$124.50 — implying, across 115 million US users, something like \$172bn of annual consumer surplus. The median was \$11.40. That eleven-fold gap is the finding: a small group of power users place enormous value on these tools, while half of all users would surrender access for a month for less than the price of a cinema ticket. The note adds, fairly, that willingness-to-accept is not willingness-to-pay, and that these figures probably *overstate* the value.

A USEFUL MENTAL MODEL

Google and Meta built the two greatest consumer businesses in history by giving the product away — because each additional user cost them almost nothing. AI chatbots invert that: a single heavy user can consume thousands of dollars of inference. Free distribution and non-trivial marginal cost are hard to hold in the same hand.

Knowledge work is where the internal research locates the only segment large enough to carry the cycle. There are perhaps 900 million to 1.3 billion digitally addressable knowledge workers. At \$30 per worker per month and full penetration, that is \$324–463bn a year. Push further — 10% of them, roughly 90 million people, paying around \$1,000 a month, which is what many enterprises already pay for premium seats — and you arrive at a \$1.08tn annual opportunity. That number clears the bar. Whether it is reachable is a different question.

READING THE CHANGE

What has actually shifted

WHAT HAS CHANGED	WHAT IT LOOKS LIKE IN THE DATA	WHY IT CUTS BOTH WAYS
The bottleneck moved from margins to demand	Internal research reversed in late July: memory-centric architecture makes inference "demonstrably profitable" on a roughly three-year payback. Yet the stocks fell.	Bullish if you think margin fears were the real overhang and are now resolved. Bearish if the market has correctly identified a harder problem the sell-side has not modelled.
Capex is now funded by debt, not just cash	Four of the five hyperscalers issued roughly \$202bn of corporate bonds in 2026 to date. Credit-default spreads on some have moved above average investment-grade levels.	Cheap capital funding a real asset is rational. But debt converts an earnings problem into a solvency question, and it makes total exposure harder to see.

Earnings quality is deteriorating	Internal research notes more than 70% of Alphabet's quarterly income came from unrealised investment gains — largely the SpaceX listing. Ex-"other income": Amazon 21.6× reported P/E to roughly 40×; Alphabet 17.4× to roughly 35×.	Unrealised gains are real gains, and Alphabet's core business is formidable. But an earnings line driven by another company's share price is not durable, and it reverses.
The financing has become circular	NVIDIA acting as a backstop of up to \$250bn on OpenAI's \$500bn data-centre programme, plus discussions of roughly \$350bn of chip financing.	Vendor financing is how capital-intensive industries have always bootstrapped. It also obscures whether demand is truly independent of the supplier.
Rates are an amplifier	The 30-year Treasury eased to 5.24% on 3 August, the highest area since 2007, while the broad index held near highs.	Long-duration profits are worth less at 5.24% and data centres cost more to finance — but rates did not cause a rotation this specific.

THE TWO CASES

Bull and bear, stated fairly

The case for the buildout

- **Unit economics have improved.** Internal research reversed a long-held sceptical position in late July, citing memory-centric hardware delivering roughly ten times the performance per watt — concluding inference is profitable on a roughly three-year payback at 65% utilisation.
- **Fundamentals are extraordinary right now.** Records at Samsung, TSMC and ASML are cash, not narrative. Palantir's Q2, reported 3 August, showed revenue of \$1.94bn, up 93%, US commercial up 149%, and raised full-year guidance.
- **Knowledge work is a big enough prize.** If even 10% of digitally addressable knowledge workers pay premium enterprise rates, the arithmetic clears the required return.
- **Forced selling is largely done.** Leveraged vehicles have been liquidated; the marginal seller is discretionary again.
- **The rotation was not a market break.** The S&P 500 held within about 1% of a record through the worst semiconductor month since 2008.

The case for caution

- **Willingness to pay looks thin.** A median of \$11.40 a month to forgo chatbots, and 94% of ChatGPT weekly users on the free tier, are hard to reconcile with a trillion-dollar revenue requirement.
- **Demand sits with very few counterparties.** Hyperscaler remaining performance obligations lean heavily on OpenAI and Anthropic — neither of which funds those commitments from operating cash.
- **Margins are structurally worse than software.** OpenAI's gross margin was reported below 40% in Q1 2026 (The Information, via internal research), against roughly 90% for traditional software.
- **Free cash flow has turned.** Alphabet went cash-flow-negative for the first time in twenty-two years while raising capex — and the market's reflex has flipped from rewarding capex to punishing it.
- **The historical analogue is unflattering.** The internal Greed & Fear view has long argued the cycle ends "not because the hyperscalers suddenly rein in their spending but because markets start to push back against that spending."

It is worth flagging one place where the internal and external views point in opposite directions. Consensus among the 21 analysts tracked on SpaceX carries an average twelve-month target of \$236.71 against a 3 August price of \$114.53 — a bullish external stance. The internal Heyokha view is the reverse: White Box calls the valuation "absolute madness," notes the company trades above firms with seven to thirty-eight times its revenue, describes the AI segment as "enormously unprofitable" after depreciation and R&D, and disclosed not participating in the listing. Both views cannot be right, and tonight's report is the first real evidence either way.

THE DASHBOARD

What to watch from here

SIGNAL	PLAIN-ENGLISH READING	WHY IT MATTERS
SpaceX Q2, tonight after the close	Consensus looks for roughly \$6.9bn of revenue, up about 68%. The segment split matters far more than the headline.	The first audited look at whether space-based and terrestrial AI compute earns a margin, or whether Starlink is subsidising it.
The 6 August lock-up expiry	The first large tranche of SpaceX shares becomes freely tradable two days after earnings.	Supply arriving into a stock already down roughly half from its June high is a live technical risk regardless of the fundamentals.
Paid-conversion disclosure	Microsoft has stopped disclosing year-on-year Copilot seat-add growth and offers no forward guidance on it.	Reduced disclosure at the exact moment investors most want the number is itself information.
Hyperscaler credit spreads	Roughly \$202bn of 2026 bond issuance; some CDS above average investment-grade levels.	Credit usually notices trouble before equity. This is the cleanest early-warning channel available.
Semiconductor equipment orders	Equipment makers fell 30–40% in July on fears about the <i>next</i> capex round, not this one.	Order books here are the market's best forward read on whether the buildout continues at pace.
Token pricing	A price war is underway, with reported cuts of 20–80% across model tiers.	Falling revenue per token against fixed infrastructure cost compresses the payback maths directly.

WAYS TO THINK ABOUT IT

Three postures, not one answer

Picks and shovels

Own the suppliers whose revenue is contracted and visible — foundry, lithography, memory — and accept the cyclicity. The internal Greed & Fear position has long favoured this over hyperscalers. The July lesson is that visible revenue does not protect a stock when the market repricing is about expectations rather than earnings.

The platform owners

Own the hyperscalers on the view that they will capture the productivity gains eventually, whatever the interim cash burn. The counter is that this is precisely where the debt sits, where free cash flow has turned, and where earnings quality has been flattered by unrealised gains.

The hedge against malinvestment

Assume some of this capital is destroyed and position for the consequence rather than the cause. The internal Greed & Fear recommendation in late July was accumulating gold and gold miners — a bet on the financing strain, not on the technology failing.

A REPRESENTATIVE MAP — these are illustrative names used to explain how the theme is expressed in equities. They are NOT recommendations, and no view on price is intended.

SPCX

SpaceX

\$114.53
3 Aug 2026

The purest and most extreme expression of the theme: a rocket and satellite monopoly that has become a vertically integrated AI compute business, with a reported \$70bn remaining-performance-obligation backlog roughly twice the size of Starlink's \$13.6bn ARR. Internal research calls the AI segment "enormously unprofitable" once depreciation and R&D are counted.

WATCH NEXT

Tonight's segment margins, and the 6 August lock-up expiry.

GOOGL

Alphabet

Q2 2026
reported

The clearest illustration of the earnings-quality question. First negative free cash flow since 2004 (–\$5.9bn) as quarterly capex doubled to \$44.9bn, with full-year guidance raised to \$195–205bn. Internal research notes more than 70% of quarterly income came from unrealised gains, largely the SpaceX listing — adjusting for which moves the P/E from about 17× to about 35×.

WATCH NEXT

Whether Q3 capex guidance holds, and whether other income reverses.

NVDA

NVIDIA

Financing
exposure

Still the toll booth on the whole buildout, but increasingly also its underwriter. Internal research flags a backstop of up to \$250bn on OpenAI's \$500bn data-centre programme plus discussions of roughly \$350bn of chip financing — the mechanism that made investors describe the demand picture as circular.

WATCH NEXT

Any expansion of customer-financing commitments, and receivables concentration.

MSFT

Microsoft

30m+ paid
Copilot seats

The cleanest read on enterprise willingness to pay. Copilot passed 30 million paid seats, up from 20 million — but that is under 7% of a base above 450 million commercial seats, and bundling means not every paid seat is paying specifically for AI. Alone among the hyperscalers, it has not issued debt in 2026.

WATCH NEXT

Whether seat-add growth disclosure is restored.

000660.KS

SK Hynix

–50% in
one month

The sharpest example of records being sold. Internal research reports net profit of \$27.8bn (up 398% year-on-year) at a 72% operating margin — with the stock down roughly 50% over the month, on a 5% revenue "miss," large non-operating gains and a heavy working-capital swing. Internal view: memory's low multiple is "only explained by market unsophistication."

WATCH NEXT

Contract memory pricing, and whether Korean retail leverage rebuilds.

PLTR

Palantir Technologies

+93% rev
Q2 2026

The demand-side counter-example, and a genuine problem for the bear case. Q2 revenue of \$1.94bn, up 93%, US commercial up 149%, adjusted EPS of \$0.41 against \$0.34 expected, \$9.2bn of cash and no debt, with full-year guidance raised to roughly \$8.15bn. Someone, evidently, is paying.

WATCH NEXT

Whether the software layer keeps compounding while infrastructure de-rates.

The bottom line

The July selloff was not the market deciding AI does not work. It was the market moving its scepticism from one part of the chain to another. For two years the worry was whether the

hardware could ever run at a profit; internal Heyokha research now argues, having reversed its own position, that it can. What replaced that worry is harder to model: whether demand at the far end of the chain is broad, durable and price-insensitive enough to service \$2–3tn of capital.

The evidence is mixed, and this edition has tried not to smooth that over. Palantir grew 93% and raised guidance yesterday. Half of chatbot users would give the technology up for \$11.40. Both are true. The reasonable inference is not that the thesis is broken but that its *distribution* of outcomes has widened — and that the market has stopped granting capex the benefit of the doubt. Tonight's SpaceX report is the first audited datapoint on the most aggressively valued expression of the trade, arriving forty-eight hours before a large tranche of its shares becomes tradable.

Editorial note. This Field Guide is educational commentary prepared for internal use at Heyokha Brothers. It is not investment advice, not a recommendation to buy or sell any security, and not personalised to any individual's circumstances. Named companies illustrate how a theme is expressed in public markets; their inclusion implies no view on their merits. External figures are attributed with as-of dates and were verified at the time of writing; markets move and figures date quickly. Internal Heyokha Wiki views are the opinions of named third-party research providers, labelled as views rather than verified forecasts; where Wiki content itself cites third parties, we have not independently confirmed those underlying claims. Where internal and external views conflict, we have said so rather than resolved it.

Sources and update notes

Internal — Heyokha Wiki. Primary page: *AI Capex* (theme, status active, created 23 June 2026, **last reviewed 4 August 2026**), sections Thesis and Contradictions. Supporting page: *SpaceX* (company, last reviewed 25 June 2026). Provider issues relied on: **White Box** (Ignacio de Gregorio Noblejas) — "Why Markets Don't Trust AI (and why I don't either)," 4 August 2026, the principal source for this edition; also 28 July 2026 ("AI is Profitable — The Real Question is How Much"), 9 July 2026 ("The Ultimate Guide for AI in 2026"), 13 June 2026 (SpaceX listing and valuation), 31 July 2026 (memory results). **Greed & Fear** (Chris Wood) — 23 July, 30 July and 2 July 2026 issues on hyperscaler credit, remaining performance obligations and free-cash-flow deterioration. **13D Research** and **Michael Burry** (23 May 2026) are cited on the page for the opposing and bearish framings respectively.

Limitation on the internal view. The page draws on multiple providers, but the current high-frequency evidence base is heavily weighted to one — White Box — which supplies most of the July–August 2026 datapoints and the demand-quality argument at the centre of this edition. Greed & Fear contributes the credit and malinvestment framing; 13D the structurally bullish counterweight. Treat the demand-side conclusions as one provider's high-conviction analysis, not internal consensus. The page carries an explicit *Contradictions* section, including the same provider reversing his own profitability view in late July 2026 — a reminder that the internal position here has been unstable.

External sources. (1) Fortune, "Wall Street's favorite bet comes undone as chips whipsaw market," 2 August 2026 — SOX –21% in July, worst month since October 2008, –28% from the 22 June record close of 14,634.72. (2) Alphabet Q2 2026 results reporting (Search Engine Journal, MLQ News), late July 2026 — free cash flow –\$5.9bn, capex \$44.9bn, full-year guidance raised to \$195–205bn. (3) Investing.com and CNBC quote pages for SPCX, as of 3 August 2026 — \$114.53 close, ~\$1.7tn market capitalisation, 16 June high of \$225.64, 28 July low of \$107.01, consensus twelve-month target \$236.71 across 21 analysts. (4) Businesswire and CNBC, 3 August 2026 — Palantir Q2 2026 revenue \$1.94bn (+93% y/y), US commercial +149%, adjusted EPS \$0.41, cash \$9.2bn, FY guidance ~\$8.15bn. (5) CNBC and Forbes previews plus Yahoo Finance, late July–3 August 2026 — SpaceX first earnings scheduled after the close on 4 August, consensus Q2 revenue ~\$6.9bn (+68%), 2025 revenue \$11.4bn and operating income \$4.4bn, Starlink 10.3m subscribers as of 31 March 2026, first major lock-up expiry 6 August. (6) Trading Economics, as of 3 August 2026 — US 30-year Treasury yield 5.24%.

Editorial choices made in this automated run. AI Capex was selected because tonight's SpaceX report is the day's largest scheduled catalyst and the Wiki frames SpaceX explicitly as an AI-infrastructure play rather than a space business — the day's news tests this thesis more directly than any other. The recency of the archive could not be checked against prior editions because the "HB Wiki Learning" folder was not connected to this session; if AI Capex has been covered recently, this edition is nonetheless anchored on new material (a source note published today).